

# Deepak Sathyan

## Articles in Review

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3. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026  
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan  
[arxiv.org/abs/2605.08010](https://arxiv.org/abs/2605.08010)
2. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025  
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan  
[arxiv.org/abs/2508.21064](https://arxiv.org/abs/2508.21064)
1. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025  
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang  
[arxiv.org/abs/2507.01000](https://arxiv.org/abs/2507.01000)

## Articles in Refereed Journals

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5. **New laboratory constraints on neutrinophilic mediators** July 2024  
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang  
[arxiv.org/abs/2407.12738](https://arxiv.org/abs/2407.12738) (Phys.Lett.B 868 (2025) 139765)
4. **‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m<sub>W</sub>** Apr 2024  
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2404.17574](https://arxiv.org/abs/2404.17574) (JHEP 02 (2025) 139)
3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023  
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2310.13687](https://arxiv.org/abs/2310.13687) (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022  
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan  
[arxiv.org/abs/2212.03929](https://arxiv.org/abs/2212.03929) (JHEP 06 (2023) 021)
1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020  
K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan  
[arxiv.org/abs/2008.06480](https://arxiv.org/abs/2008.06480) (JHEP 11 (2020) 109)

## Articles in Preparation

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1. **LHC as a Beam Dump: Probing Light Dark Photons** Sept 2026  
B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan  
[arxiv.org/a/sathyan\\_d\\_1](https://arxiv.org/a/sathyan_d_1)

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| <p><b>2. New Probes of Dark Sectors at a Muon Collider</b><br/>         B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan<br/> <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a></p>                                                          | <p>Jan 2027</p> |
| <p><b>3. Exploring open systems approach to neutrino oscillations</b><br/>         A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou<br/> <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a></p>                                        | <p>Mar 2027</p> |
| <p><b>4. LHC Signals for KK Gravitons Producing 8-Particle Final States</b><br/>         K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan<br/> <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a></p> | <p>Mar 2027</p> |

## Ph.D. Thesis

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| <p><b>1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass</b><br/>         D. Sathyan<br/> <a href="https://drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9">drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9</a> (Ph.D. Thesis)</p> | <p>2024</p> |
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## Snowmass2021 Contributions

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| <p><b>5. TF07 Snowmass Report: Theory of Collider Phenomena</b><br/>         F. Maltoni et al.<br/> <a href="https://arxiv.org/abs/2210.02591">arxiv.org/abs/2210.02591</a></p>                                                                                                                           | <p>Oct 2022</p>  |
| <p><b>4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021</b><br/>         K. Agashe et al.<br/> <a href="https://arxiv.org/abs/2209.11267">arxiv.org/abs/2209.11267</a></p>                                                                                | <p>Sept 2022</p> |
| <p><b>3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021</b><br/>         T. Bose et al.<br/> <a href="https://arxiv.org/abs/2209.13128">arxiv.org/abs/2209.13128</a></p>                                                                           | <p>Sept 2022</p> |
| <p><b>2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond</b><br/>         K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan<br/> <a href="https://arxiv.org/abs/2204.02928">arxiv.org/abs/2204.02928</a></p>           | <p>Apr 2022</p>  |
| <p><b>1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models</b><br/>         K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan<br/> <a href="https://arxiv.org/abs/2203.13305">arxiv.org/abs/2203.13305</a></p> | <p>Mar 2022</p>  |

## Muon g-2 Articles

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3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** Nov 2021  
B.T. King et al.  
[arxiv.org/abs/2111.02076](https://arxiv.org/abs/2111.02076) (JINST 17 (2022) P02035)
2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021  
Muon g-2 Collaboration  
[arxiv.org/abs/2104.03240](https://arxiv.org/abs/2104.03240) (Phys.Rev.Accel.Beams 24 (2021) 044002)
1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** Apr 2021  
Muon g-2 Collaboration  
[arxiv.org/abs/2104.03281](https://arxiv.org/abs/2104.03281) (Phys.Rev.Lett. 126 (2021) 141801)